

Latency Vs Throughput

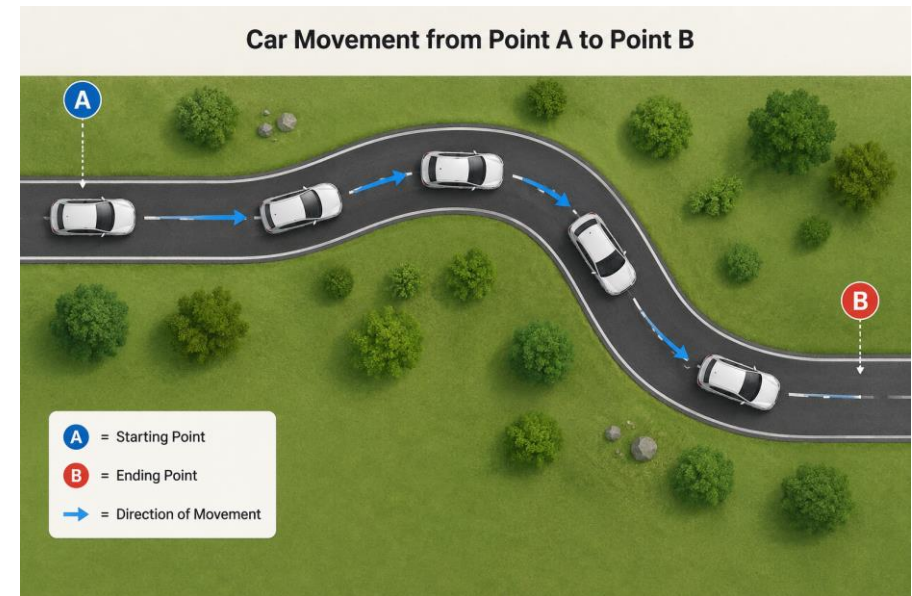
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Latency

- **Latency** measures **speed**—

The time delay for a single request to travel from source to destination.

- Latency is measured in **milliseconds (ms)**.
- Common tools used to measure latency include:
 - **Ping**
 - Network diagnostic tools
 - Traceroute (to identify delay across hops).

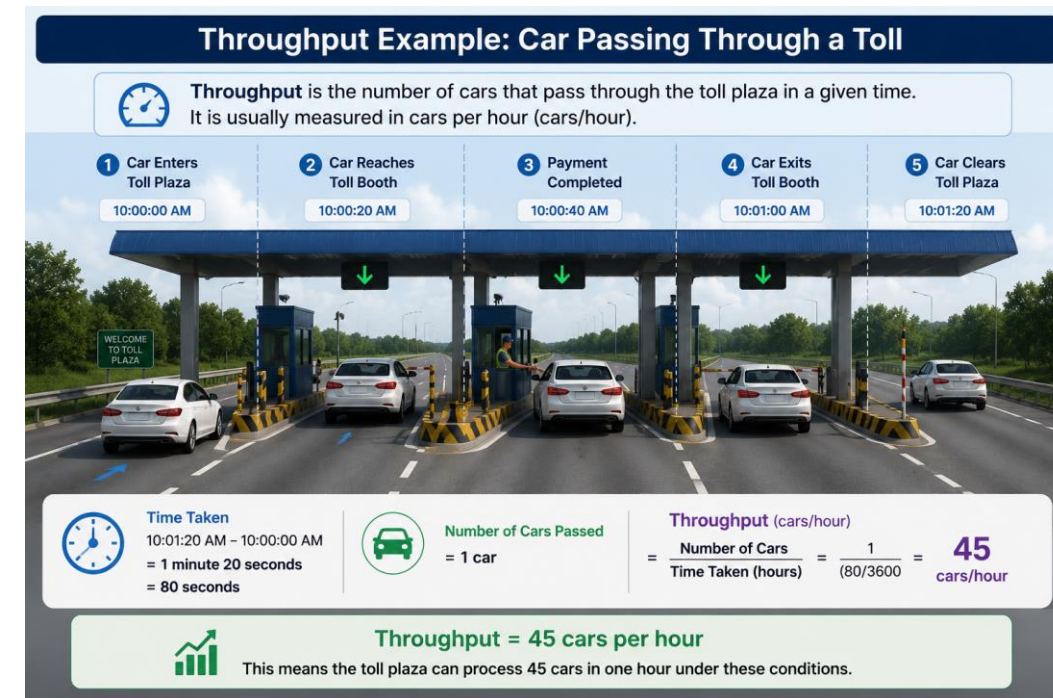


Throughput

- **Throughput** measures **capacity**—
The total amount of data or tasks successfully processed over a given period.
- Throughput is measured in bits per second (bps), most commonly:
 - Mbps
 - Gbps

It is measured using:

- Network traffic generators
- File transfer tests
- Monitoring tools that track data flow rates



Key Differences

Feature	Latency	Throughput
What it measures	Delay / Response time	Volume / Processing capacity
Common Unit	Milliseconds (ms)	Bits per second (bps) or requests/sec
Why it matters	Critical for real-time apps such as: Online meetings Video calls Online gaming Financial trading systems	Critical for batch processing, video streaming, and large downloads.
Optimized by	Shortening physical distance and reducing network hops.	Adding parallel processing, load balancers, or wider bandwidth.

Relationship Between Latency and Throughput

- Latency and throughput are **related but independent**
- A system can have **low latency but low throughput** (fast responses but limited users).
- A system can have **high throughput but high latency** (handles many users, but each waits longer).
- Well-designed systems **aim to achieve low latency and high throughput together**.

Why Both Matter in System Design

- **Latency** is critical for user-facing applications like search engines, payment systems, and real-time apps.
- **Examples of user-facing applications**
 - Social media apps like Instagram and WhatsApp
 - E-commerce websites like Amazon
 - Streaming apps like Netflix
 - Banking apps
 - Food delivery apps
 - Ride-booking apps
 - Online learning platforms
- **Throughput** is crucial for backend systems, batch processing, and high-traffic platforms.

In a toll system:

The app showing toll balance to drivers
→ **user-facing**

The server calculating throughput and storing vehicle data → **backend system**

• **Latency Comparison Numbers (~2012)**

• L1 cache reference	0.5 ns	
• Branch mispredict	5 ns	
• L2 cache reference	7 ns	14x L1 cache
• Mutex lock/unlock	25 ns	
• Main memory reference	100 ns	20x L2 cache, 200x L1 cache
• Compress 1K bytes with Zippy	3,000 ns	3 us
• Send 1K bytes over 1 Gbps network	10,000 ns	10 us
• Read 4K randomly from SSD	150,000 ns	150 us ~1GB/sec SSD

- Read 1 MB sequentially from memory 250,000 ns 250 us
- Round trip within same datacenter 500,000 ns 500 us
- Read 1 MB sequentially from SSD 1,000,000 ns 1,000 us 1 ms ~1GB/sec
SSD, 4X memory
- Disk seek 10,000,000 ns 10,000 us 10 ms 20x datacenter roundtrip
- Read 1 MB sequentially from disk 20,000,000 ns 20,000 us 20 ms 80x
memory, 20X SSD
- Send packet CA->Netherlands->CA 150,000,000 ns 150,000 us 150 ms

- Notes

- 1 ns = 10^{-9} seconds
- 1 us = 10^{-6} seconds = 1,000 ns
- 1 ms = 10^{-3} seconds = 1,000 us = 1,000,000 ns

Thank You